

CURRICULUM VITAE



Dr. SEKAR TAMILPERUVALATHAN

Associate Professor & Head

P.G-Manufacturing Engineering

Department of Mechanical Engineering

Government College of Technology (GCT)

Coimbatore-641 013, Tamil Nadu, India

Mobile: +91-9092500393



E-mail: drtsekar76@gct.ac.in & tsekargct@gmail.com

College Profile: <https://gct.ac.in/users/sekar>

Google Scholar link:

https://scholar.google.co.in/citations?hl=en&user=mTT1jrAAAAAJ&view_op=list_works&sortby=pubdate

Scopus ID: 56152897600

Scopus Link:

<https://www.scopus.com/authid/detail.uri?authorId=56152897600>

Web of Science Link:

<https://www.webofscience.com/wos/author/record/JOK-2774-2023>

Web of Science ID: JOK-2774-2023

LinkedIn Link:

www.linkedin.com/in/sekar-t-27a707116

Teaching, Research, Industry and Administrative Experience in Engineering Colleges:

27 Years 9 Months

(Under Tamil Nadu Educational Services, Government of Tamil Nadu since 2001)

GCT- Officer –in-charge of Examinations

GCT-Intellectual Property Rights (IPR) Cell Coordinator

Number of Design Registration Certificates from Government of India (Design Patents)

Obtained: 10

Educational Qualifications:

1. **Ph.D.** in Mechanical Engineering, **Anna University, Chennai**- July, 2009.
2. **M.E.** in Production Engineering (*Gold Medalist*), **Annamalai University, Chidambaram** - November, 1999.
3. **B.E.** in Mechanical Engineering, **PSG College of Technology, Coimbatore** - April, 1997.

4. **Chartered Engineer** – Institution of **Mechanical Engineer (IMechE)** -United Kingdom- Recognized by **British Royal Council** (From October, 2019 to December, 2022).
5. **NEBOSH** (Industrial Safety- Certificate Level)
6. **M.B.A.** in Human Resources, **Periyar University, Salem** - July, 2014.
7. **3D Printing and Design**- Trained by **Indian Institute of Technology, Hyderabad**.

Professional Experience:

1. Govt. Engineering Colleges under Tamil Nadu Educational Services/

Abroad Service/ Self-financing Engineering Colleges

(Excluding Full-Time M.E)

: 27 Years 3 Months

2. Work experience in Die-Casting Industry, Visakhapatnam

: 6 Months

Summary of Publications:

1. **International Journals** : **90**
2. National Journals : 02
3. International Conferences : 100
4. National Conferences : 40
5. Books Authored : 04
6. Book chapter published : 02
7. U.G/P.G Projects Guided : 75
8. **Ph.D Guided** : **06**
9. Ph.D Guiding : 05
10. **Number of Design Certificates Granted by Indian Patent Office: 10**
11. Number of Utility Patents Filed: 01

Recently published International Journals:

1. Daniel Antony Arokiyasamy and **Sekar Tamilperuvalathan** (2026), ‘Next-Generation PLA Based Composites: Structure–Property Application Correlation with Sustainable Reinforcement Systems’. Polymer Science Series C, Springer Publications <https://doi.org/10.1134/S1811238225600235>
2. Daniel Antony Arockiyasamy and **Sekar Tamilperuvalathan** (2026), ‘A comprehensive evaluation of material properties, degradation kinetics and clinical performance of biodegradable polymer cardiovascular stents: A review’, International Journal of Biological Macromolecules. Elsevier, (Q1, IF:8.5) Article ID: 150358. <https://doi.org/10.1016/j.ijbiomac.2026.150358>.

3. **Sekar Tamilperuvalathan**, Vinoth Varadharaju, Sakthivel Rajamohan, Dhinesh Balasubramanian, Utku Kale & Artūras Kilikevičius (2026). 'Performance enhancement of electrochemical discharge micromachining of borosilicate glass using nitrogen gas assistance', Science Reports, Springer Nature, 16, 18700. <https://doi.org/10.1038/s41598-026-57797-4>
4. Daniel Antony Arockiyasamy and **Sekar Tamilperuvalathan** (2025), 'Bio-functional Evaluation of FDM 3D Printed Poly(lactic acid) /Ormocarpum Cochinchinense Bio-Composites for Mechanical Performance, Microbial Inhibition and Wound Healing Efficiency', International Journal of Biological Macromolecules. Elsevier, (Q1, IF:8.5). <https://www.sciencedirect.com/science/article/abs/pii/S0141813025105540>
5. **Sekar Tamilperuvalathan**, Vinoth Varadharaju, Vijay Manoharan, Sakthivel Rajamohan, Dhinesh Balasubramanian, Utku Kale, Artūras Kilikevičius and Vilma Locaitienė (2025), 'Enhancing electrochemical discharge drilling performance in mild steel sheet using KMnO₄ mixed electrolytes', Scientific Reports, 15, Article No. 39703. (Q1 with IF:4.3) <https://doi.org/10.1038/s41598-025-23399-93>.
6. Venkatesh Chenrayan, Panneerselvam Natarajan, **Sekar Tamilperuvalathan**, Kiran Shahapurkar and HC Ananda Murthy (2025), 'Enhancing flexural performance of friction-stir-processed composites via SiC and Y₂O₃ reinforcements: a combined experimental and numerical study', Emergent Materials, Springer Publications, (Q2 with IF: 3.8) <https://doi.org/10.1007/s42247-024-00650-z>.
7. **Sekar Tamilperuvalathan** and Daniel Antony Arockiyasamy (2025), 'Polylactic acid/Ormocarpum cochinchinense bio-composites for sustainable 3D-printed biomedical devices', 3 Biotech (Springer) 15, 373. (Q1 with IF:2.9) <https://doi.org/10.1007/s13205-025-04550-9>.
8. Panneerselvam Natarajan, **Sekar T**, Venkatesh Chenrayan and L. Rajeshkumar (2025), 'The effect of SiC and Y₂O₃ inclusion on microstructure and mechanical properties of Al 5052 composite fabricated through Friction Stir Process', Heliyon (11) e4166, Cell Press, Elsevier Publication (Q1 with IF:3.4) <https://doi.org/10.1016/j.heliyon.2025.e41665>.
9. **T.Sekar**, U.R.Nivetha, R.Jayashree, R.Surendran, and A.Daniel Antony (2025), 'Compressive Strength Evaluation of Poly(Lactic Acid) and PLA blended with Blepharis Maderaspatensis for Orthopedic Applications', Indian Industrial

Engineering Journal, Vol.54, Issue.1, No.1, pp.96-101. ISSN: 0970-2555 (UGC CARE Approved Group-I Peer Reviewed Journal)

10. **T.Sekar**, A.Hariprabu, R.Surendran, R.Jayashree and A.Daniel Antony (2025), 'Evaluation of Biocompatibility and Antimicrobial properties of Bio-Composites for Kneecap Applications', Indian Industrial Engineering Journal, Vol.54, Issue.1, No.2, pp.152-156. ISSN: 0970-2555 (UGC CARE Approved Group-I Peer Reviewed Journal)
11. R.Jayashree, R.Surendran, S.Vijay and **T.Sekar** (2025), 'Investigation on the performance of IoT based Electrochemical Discharge Machining', Indian Industrial Engineering Journal, Vol.54, Issue.1, No.3, pp.1-5. ISSN: 0970-2555 (UGC CARE Approved Group-I Peer Reviewed Journal)
12. R.Jayashree, S.Thilothama, **T.Sekar** and R.Surendran (2025), 'Design, Fabrication and Analysis of a Suspension fork cap wrench socket using CNC Machining and Additive Manufacturing', Indian Industrial Engineering Journal, Vol.54, Issue.2, No.1, pp.13-15. ISSN: 0970-2555 (UGC CARE Approved Group-I Peer Reviewed Journal)
13. R.Surendran, G.P.Beslin Bebish, **T.Sekar**, R. Jayashree and N.Nandakumar (2025), 'Prioritizing barriers in Thermal Power Plant using Fuzzy Logic: A Sustainable Perspective', Indian Industrial Engineering Journal, Vol.54, Issue.1, No.1, pp. 79-84. ISSN: 0970-2555 (UGC CARE Approved Group-I Peer Reviewed Journal)
14. N.Nandakumar, S.Sargunam, **T.Sekar**, R.Surendran and R.Jayashree (2025), 'Experimental Investigation of Mechanical Properties of 3D Printed PLA/TPU/CAPRA Hircus Powder Composites', Indian Industrial Engineering Journal, Vol.54, Issue.1, No.1, pp.154-160. ISSN: 0970-2555 (UGC CARE Approved Group-I Peer Reviewed Journal)
15. Elango Natarajan, **Sekar Tamilperuvalathan**, Saravanakumar Nesappan, Anto Dilip Albert Selvaraj and Santhosh Mozhuguan Sekar (2024), 'Structural Integrity and Mechanical behavior of Environmentally Friendly Filler as reinforcement in PLA-HDPE Composite', Emergent Materials, Springer Publications, IF: 3.8, Published online on 01.03.2024. <https://doi.org/10.1007/s42247-024-00650-z>.
16. Vijay Manokaran, **Sekar Tamilperuvalathan**, Prasanth Ponnusamy and Elango Natarajan (2024), 'Exploring Nitrogen Gas-Assisted ECD μ M of Glass by Modified Blackwidow Optimization', Materials Today Communications, Elsevier publications,

Vol. 38, 108295, pp.1-14. IF:3.8. (doi: [https:// doi.org/10.1016/ j.mtcomm. 2024. 108295](https://doi.org/10.1016/j.mtcomm.2024.108295)).

17. Panneerselvam Natarajan, **Sekar Tamilperuvalathan**, Vijayakumar Murugesan and Kumaresan Govindasamy (2024), 'An enhanced technique for friction stir welding of ceramic particle reinforced Aluminium based metal matrix composites', *Lubrication Science*, Wiley Publisher, Vol.36, Issue 3, pp. 162-176 IF: 1.302 (DOI: [https:// doi.org/ 10.1002/lis.1687](https://doi.org/10.1002/lis.1687))
18. G K Barath, **T Sekar**, N Nandhakumar, S Periyasamy and Daniel Antony Arokiyasamy (2024), 'Mechanical Testing of 3D-Printed PLA specimens with different infill patterns for healthcare applications', *Indian Institution of Industrial Engineering* Vol.53, Issue.12, No.3, pp.191-196. ISSN: 0970-2555 (UGC CARE Approved Group-I Peer Reviewed Journal)
19. Adithyaa A, **Dr. T. Sekar** and Daniel Antony A (2024), 'Investigation of material characteristics of Hybrid Biopolymers for Ankle Knee Replacement', *Science, Technology and Development Journal UGC CARE Group -2 Journal*, Vol. XIII, Issue 6, pp. 19-40 (DOI:24.18001. AJCT. 2024. V17I6.24.6102) (ISSN : 0950-0707)
20. Ashwin Kumar S, Surendran R and **Dr T Sekar** (2024), 'Optimization of dry sliding wear behavior of ALSi10MG alloy produced by laser-based powder bed fusion using Response Surface Methodology', *Science, Technology and Development Journal UGC CARE Group -2 Journal*, Vol. XIII, Issue 6, pp. 133-168 (DOI:24.18001.AJCT.2024.V17I6.24.6118) (ISSN : 0950-0707)
21. Nathish Sanjay B N, **Sekar T**, Daniel Antony Arokiyasamy (2024), 'Structural failure analysis of the high-speed shaft coupling arm in Wind turbine', *Indian Institution of Industrial Engineering* Vol.53, Issue.6, No.2, pp.19-27. ISSN: 0970-2555 (UGC CARE Approved Group-I Peer Reviewed Journal)
22. Rahul S R, **Sekar T**, Daniel Antony Arokiyasamy (2024), 'Investigations on the Design of Bio-Degradable Cardiovascular Stent in Healthcare Applications', *Journal, Indian Institution of Industrial Engineering*, Vol.53, Issue.4, pp.130-140. ISSN: 0970-2555 (UGC CARE Approved Group-I Peer Reviewed Journal)
23. Rahul S R, **Sekar T**, Daniel Antony Arokiyasamy (2024), 'Bio-Degradable Polymeric Stents : A focussed Review on Materials and Clinical Studies', *International Journal of Reserach Publication and Reviews*, Vol.5, No.2, pp.795-800. ISSN: 2582-7421

24. M.Vijay and **Sekar Tamilperuvalathan** (2023), 'Experimental investigation and prediction of ECDM parameters on fiber reinforced SiC composite using hybrid ERNN-based Sparrow Search Optimization', *Materials Today Communications*, Elsevier publications, Volume 36, 106777. IF:3.8 (doi: <https://doi.org/10.1016/j.mtcomm.2023.106777>).
25. **Sekar Tamilperuvalathan**, E. Natarajan, C. K. Ang, L. W. Hong and P. Rajendran (2023) "Modeling of Rotary Tool Adapted Electrochemical Machining of AISI 202," IEEE 19th International Conference on Automation Science and Engineering (CASE), Auckland, New Zealand, pp. 1-7, (doi: 10.1109/CASE56687.2023.10260464)
26. P Ponnusamy, **Sekar Tamilperuvalathan** (2023), 'Performance evaluation and hybrid deep recurrent neural network-based prediction of SS304 turning characteristics using nanoparticles added water emulsified MQL', *Biomass Conversion and Bio-refinery*, Springer Publication, Vol.13, pp.7349-7373. IF: 4.05. <https://doi.org/10.1007/s13399-023-04106-y>.
27. Rahul S R, **Sekar T**, Daniel Antony Arokiyasamy (2023), 'Design and Analysis of Bio-Degradable Cardiovascular stent in healthcare Applications', *Journal of Engineering Technology and Management Sciences*, Vol.7, Issue 5, pp.360-367 (DOI: 10.46647 /ijetms. 2023.v07i05.042, ISSN: 2581-4621)
28. Rahul S R, **Sekar T**, Nandakumar N and Periyasamy S (2023), 'Design and Analysis of Cardia Stent', *International Journal of Engineering Technology and Management Sciences*, Vol.7, Issue 5, pp.154-161 (DOI: 10.46647 /ijetms. 2023.v07i05.018, ISSN: 2581-4621)
29. V Vinoth, **Sekar Tamilperuvalathan** and M Kumaran (2023), 'Investigations on the Mechanical Characteristics of the Stainless Steel 316L Alloy Fabricated by Directed Energy Deposition for Repairing Application', *Journal of Materials Engineering and Performance*, Springer Publication, Vol 32, pp.4138–4150. IF: 2.3. <https://doi.org/10.1007/s11665-023-07884-8>.
30. **Sekar Tamilperuvalathan**, Haritha Kaliyaperumal Rabindranathan , Rajeswari Balakrishnan , Prabakaran Jayaraman, Surendran Ramakrishnan (2023), ' A Critical Review of the impacts of 3D Printing Technologies on Dental Medicine', *International Journal of Engineering Research and Technology*, Vol.12, Issue 4, pp:306-323.

31. Sakthi Kumar Arumugam, **Sekar Tamilperuvalathan**, Daniel Antony Arokiyasamy, Gnanaprakash Kanagaraj and Surendran Ramakrishnan (2023), 'Electrochemical discharge machining studies on HSS milling cutter', International Journal of Progressive Research in Engineering Management and Science (IJPREMS), Vol. 03, Issue 07, pp.336-341.
32. Gnanaprakash Kanagaraj, **Sekar Tamilperuvalathan**, Daniel Antony Arokiyasamy, Sakthi Kumar Arumugam and Surendran Ramakrishnan (2023), 'Synthesis of Rigid Polyurethane foams with Palm Oil based flame retardant polyols, International Journal of Progressive Research in Engineering Management and Science (IJPREMS), Vol. 03, Issue 07, pp.449-459. (https://www.ijprems.com/uploadedfiles/paper/issue_7_july_2023/31848/final/fin_ijprems1690787287.pdf) e-ISSN: 2583-1062
33. V.Priyadarshan, R.Surendran, **Dr.T.Sekar**, and D.Ajith kumar (2023), 'Experimental Investigation on Cutting Force Analysis in LM6 Aluminium Nano-Composite Material', International Journal of Innovative Science and Research Technology, Vol. 8, Issue 7, pp.3281-3284. ISSN No:-2456-2165 (<https://ijisrt.com/assets/upload/files/IJISRT 23JUL1225.pdf>)
34. D. Ajith kumar, Dr. B. Rajeswari, **Dr.T.Sekar**, R.Surendran, V.Priyadarshan and A.Sakthikumar (2023), 'Preparation of 3D Printing Filament from Face Mask Waste', International Journal of Innovative Science and Research Technology, Vol. 8, Issue 7, pp.449-459. ISSN No:-2456-2165 (<https://www.ijisrt.com/assets/upload/files/IJISRT23JUL1865.pdf>)
35. **Sekar Tamilperuvalathan**, Haritha Kaliyaperumal Rabindranathan , Rajeswari Balakrishnan , Prabakaran Jayaraman, Surendran Ramakrishnan (2023), ' A Critical Review of the impacts of 3D Printing Technologies on Dental Medicine', International Journal of Engineering Research and Technology, Vol.12, Issue 4, pp:306-323. DOI : 10.17577/IJERTV12IS040148
36. Vijayakumar Kanniyappan and **Sekar Tamilperuvalathan** (2022), "Investigation and Prediction of ECMM Characteristics of hardened die steel with Nano particles added Electrolytes using Hybrid Deep Neural Network', Polish Journal of Chemical Technology, Vol.24, Issue 4, pp. 7-22, e-ISSN:1899-4741<https://doi.org/10.2478/pjct-2022-0024>

37. M.Veluchamy, R.Surendran and **T.Sekar** (2022), "Experimental Investigation on Testing of Al 7085/Alumina Nano Composites by using Stir Casting Method', International Research Journal of Engineering & Technology, Vol.9, Issue 7, pp. 423-28, e-ISSN:2395-0056
38. T.Arunkumar, B.Rajeswari and **T.Sekar** (2022), "Study on Gas Metal Arc Welding in S235 Steel', International Research Journal of Engineering & Technology, Vol.9, Issue 6, pp. 2811-2814, e-ISSN:2395-0056
39. P.Kumaresanan, S.Periyasamy, **T.Sekar** and N.Nandakumar (2022), 'Experimental Investigation of PVC Composite Membrane for Polymer Electrolyte Membrane Fuel Cell', International Advanced Research Journal in Science, Engineering and Technology, Vol. 9, Issue 6, pp.916-921 ISSN (O) 2393-8021, ISSN (P) 2394-1588 DOI: 10.17148/IARJSET.2022.96149
40. V.R.R.Dilibban, N.Nandakumar and **T.Sekar** (2022), 'Experimental Analysis of Orthopedic Staple Pin made by Nickel Titanium Alloy', International Research Journal of Engineering & Technology, Vol.9, Issue 6, pp. 2768-2772, e-ISSN:2395-0056
41. B.Vishal Easwaran, **T.Sekar**, N.Nandakumar and B.Rajeswari (2022), "Development of Fourth Link and its impact on FDM 3D Printer' International Advanced Research Journal in Science, Engineering and Technology (IARJSET), Vol.9, Issue 7, pp. 49-58. DOI: 10.17148/IARJSET.2022.9709 & ISSN (O) 2393-8021, ISSN (P) 2394-1588
42. N.Elango, V.Kaviyarasan, Wei Hong Lim, S.Ramesh, **T.Sekar** and V.H.Mok (2022), 'Gorilla Troops Optimizer Combined with ANFIS for Wire Cut EDM of Aluminum Alloy', Advances in Materials Science and Engineering, Hindawi, Vol. 2022, Article ID 3072663, 14 pages <https://doi.org/10.1155/2022/3072663>.
43. A.R.Udhaya, **T.Sekar**, N.Nandakumar and R.Surendran (2022), 'Experimental Investigation on Mechanical Behavior of Bio-Composite Material', International Advanced Research Journal in Science, Engineering and Technology, Vol. 9, Issue 6, pp. 203-211, ISSN (O) 2393-8021, ISSN (P) 2394-1588 DOI: 10.17148/IARJSET.2022.962
44. V.R.R.Dilibban, N.Nandakumar and **T.Sekar** (2022), 'Analysis of Thermal Energy storing and self-cleaning coating for Solar panel by using Titanium Dioxide Melamine Formaldehyde', International Research Journal of Modernization in

Engineering Technology and Science, Vol.4, Issue 6, pp.728-732, e-ISSN: 2582-5208. June Issue.

45. P. Kumaravel, P. Suresh, K. Venkatesh Raja, **T.Sekar** (2022), 'Improvement of Micro-Electrochemical Discharge Machining of Austenitic Stainless Steel 316L using NaOH electrolyte containing N₂', International Journal of Electrochemical Sciences, Vol. 17, Article Number: 220747, doi: 10.20964/2022.07.53.
46. Vinodhkumar R, **Sekar T** and Nandakumar N, (2022), 'Design and Analysis of an Extruder for Bio-Composite Filament Fabrication', Journal of Electronics and Informatics, Vol. 4, Issue 1, pp.42-57. doi:10.36548/jei.2022.1.005
47. M.Vijay and **Sekar T** (2022), 'Prediction on enhanced Electrochemical Discharge Machining behaviors of Zirconia-Silicon Nitride using Hybrid DNN based 'Spotted Hyena Optimization'', International Journal of Energy Research, Wiley Publications, Volume 46, Issue 7, Pages: 9221-9241, <https://doi.org/10.1002/er.7797>.
48. M.Vijay, **Sekar T**, Elango Natarajan and Prasanth Ponnusamy (2021), 'Experimental Investigation on Electrochemical Discharge Machining of Zirconia', Recent Advances in Manufacturing, Automation, Design and Energy Technologies, Springer Lecture Notes, pp. 27-34 ISSN:Print: 2300-1917. https://doi.org/10.1007/978-981-16-4222-7_4
49. K.Vijayakumar, **Sekar T**, and M.Vijay (2021), 'Experimental Studies on Material Removal Rate of Die Steel in Electrochemical Micromachining Process Using Taguchi Method', Recent Advances in Manufacturing, Automation, Design and Energy Technologies, Springer Lecture Notes, pp. 181-187 ISSN:Print: 2300-1917. https://doi.org/10.1007/978-981-16-4222-7_4
50. P.Prasanth, **Sekar T** and Sivapragash M (2021), 'Investigations on the effects of Nitrogen Gas in CNC Machining of SS304 using Taguchi and Firefly algorithm', Bulletin of the Polish Academy of Sciences, Vol.69. No.1 ISSN: 2300-1917.DOI: 10.24425/bpasts.2020.136211
51. Vignesh S, Surendran R, **Sekar T** and Rajeswari B (2021), 'Ballistic Performance Simulation of Graphene–Dyneema Multi-layered Armor', Materials, Design and Manufacturing for Sustainable Environment, Springer-Lecture Notes in Engineering, pp 165-173 .
52. B.Hemalatha, B.Rajeswari, **Sekar T** and V.Rajasekar (2021), 'Experimental Studies on Biomachining of Copper and its behavioral Characteristics', Materials, Design and

Manufacturing for sustainable Environment, Springer Lecture Notes, doi.org/10.1007/978-981-15-9809-8_4, pp 49-56, published in online on 07.02.2021, , Online ISSN: 978-981-15-9809-8.

53. **Sekar T**, B.Hemalatha, B.Rajeswari, R.Surendran and M.Vijay (2021), 'Investigation on the effects of Aspergillus niger in sustainable bio-micromachining of Copper', Materials Today: Proceedings, Elsevier, ISSN:2214- 7853, <https://doi.org/10.1016/j.matpr.2021.02.011>.
54. Kannan P R, Senthilkumar P, Suresh P and **Sekar T** (2020), 'Fuzzy Sliding Mode Control for Hydraulic activated active suspension using Half Car Model', International Journal of Mechanical and Production Engineering Research and Development (IJMPERD), Trans Stellar (UGC Approved), Vol. 10, Issue 4 (August), pp: 93-102.ISSN (P):-2249-6980, ISSN (E):2249-8001.
55. Mohansendrayan A, Surendran R, **Sekar T** and Nandakumar N (2020), 'Investigation on Mechanical Properties of LM13 Aluminium Hybrid Composites', Journal of Xidian University, China, Vol.14, Issue 6, pp. 2061-2069 (UGC approved Journal-June 2020 Published -ISSN: 1001-2400)
56. Senthilraja, K ,Kannan P R,Suresh P and **Sekar T** (2020), 'Corrosion and Fatigue study on FSW and GTAW welded joints of AA7075-T6', Journal of Xidian University, China, Vol.14, Issue 5, pp. 5484-5492 (UGC approved Journal- May 2020 Published -ISSN: 1001-2400)
57. Vignesh S, Surendran R, **Sekar T** and Rajeswari B (2020), 'Ballistic impact analysis of Graphene Nano-sheets reinforced Kevlar-29', Materials Today: Proceedings, Elsevier, ISSN:2214- 7853, Article In press (March, 2020)
58. Muthukumaran N, Devaraj P, Arunkumar P and **Sekar T** (2020), 'Re-Use of abandoned Sanitary ware waste as Abrasive particles for Abrasive Jet Machine', Materials Today: Proceedings, Elsevier, ISSN:2214- 7853, Article In press (March, 2020).
59. Muthukumaran N, Prasanna Raj Yadav, Saravanan C G S and **Sekar T** (2020), 'Synthesis of cracked Mahua oil using coal ash catalyst for Diesel Engine application', International Journal of Ambient Energy, Taylor and Francis, Vo. 41, Issue 3, pp. 241-256. (Print ISSN: 0143-0750 & Online ISSN: 2162-8246)

60. Sivakumar A, Sathiyamoorthy V, **Sekar T** and Seenivasan M (2019), 'Investigations on the effects of Copper Nano Fluid in Electrochemical Machining of HCHCr Die Steel', Caribbean Journal of Science, Vo. 53, Issue 1, pp. 447-453. (ISSN: 0008-6452)
61. Sivakumar A, **Sekar T** and Sathiyamoorthy V, (2019), 'Optimization of Material Removal Rate in Electrochemical Machining of AISI 202 with Mixed Electrolytes', Caribbean Journal of Science, Vo. 53, Issue 2, pp. 1352-1360. (ISSN: 0008-6452)
62. **Sekar T**, Vijay M, Rajamurugan T V and Umanath K (2018), 'Experimental Investigation on the impact of Silver Nitrate solution mixed electrolyte in the ECM of AISI 202', International Journal of Mechanical and Production Engineering Research and Development (IJMPERD), Trans Stellar (UGC Approved), Vol. 8, Special issue. 7, pp: 985-999. ISSN (P):-2249-6980, ISSN (E):2249-8001.
63. Sivakumar A, Arularasu M and **Sekar T** (2018) 'Electrochemical Machining of AISI 202 using AgNO₃ mixed Electrolyte and its Optimization', International Journal of Engineering Development and Research, Vol.6, Issue 3, pp:499-504. (ISSN: 2321-9939)
64. **Sekar T**, Vijay M, Arvindhkumar S, Boopathi S and Pavitra K (2018), 'Experimental Investigation and Optimization on Hybrid Metal Matrix Composites through Electrochemical Micro Machining using Taguchi Method', International Research Journal in Advanced Engineering and Technology, Vol.4, Issue 2, pp: 3034-3045. (ISSN: 2455-0876)
65. **Sekar T**, Arularasu M and Sathiyamoorthy V (2016), 'Investigations on the effects of Nano-fluid in ECM of Die Steel', Measurement, Elsevier, Vol. 83, pp: 38-43. (ISSN: 0263-2241)
66. Sathiyamoorthy V and **Sekar T** (2016), 'Optimization of processing parameters in ECM of AISI 202 using Multi Objective Genetic Algorithm', International Journal of Enterprise Network Management, Inderscience, Vol. 7, Issue: 2, pp: 133-141. (ISSN (E): 1748-1260)
67. Sathiyamoorthy V, **Sekar T** and Subramanian M (2016), 'Experimental investigation on the effects of Nano-fluid in ECM of High Carbon High Chromium Die Tool Steel', Asian Journal of Research in Social Sciences and Humanities, Vol.6. Issue.8, pp:100-110. (ISSN (E): 2249-7315)

68. Kannan P R, Palaniswamy E, **Sekar T**, Suresh P and Senthilkumar C (2016), ' Nano-Powder mixed ECM of Inconel 718 using ANFIS', International Journal of Advanced Engineering Technology, Vol. VII, Issue. II , pp.546-552. (ISSN (E): 0976-3945)
69. Kannan P R, Palanisamy E, **Sekar T** and Suresh P (2016), 'Prediction of Surface roughness model for Nano Copper suspended Electrochemically machined Inconel 718 using ANFIS', Asian Journal of Research in Social Sciences and Humanities, Vol.6. Issue.6 (Special), pp:104-113. (ISSN (E): 2249-7315)
70. Kumar A Siva, Arularasu M, **Sekar T** and Suresh P (2016), 'Investigation on the effects of Silver Nitrate solution mixed electrolyte in Electrochemical Machining of AISI 202', Asian Journal of Research in Social Sciences and Humanities, Vol.6. Issue.8, pp: 2019-2026. (ISSN (E): 2249-7315)
71. Suresh P, Venkatesan R, **Sekar T** and Kannan P R (2016), 'Investigation and Multi Objective Optimization of MRR and TWR using Taguchi Technique in Micro Electro Discharge Machining', Asian Journal of Research in Social Sciences and Humanities, Vol.6. Issue.6, pp: 2165-2174. (ISSN (E): 2249-7315)
72. Sathiyamoorthy V, **Sekar T** and Elango N (2015), 'Optimization of processing parameters in ECM of die tool steel using Nano fluid by Multi Objective Genetic Algorithm', The Scientific World Journal, Volume 2015 (2015), pp: 1-6. (ISSN (E): 1537-744X)
73. Sathiyamoorthy V, **Sekar T** and Subramanian M (2015), 'Optimization of Copper Nano-Fluid Used ECM By Firefly Algorithm', International Journal of Applied Engineering Research (IJAER), Vol. 10, No. 9, pp: 24287-24298. (ISSN (E): 0973-9769)
74. Suresh P, Venkatesan R, **Sekar T** (2015) 'Development of low cost 3 axes Table top Micro Electro Discharge Machining (MicroEDM)', International Journal of Applied Engineering Research (IJAER), Vol. 10, No.50, pp:226-229. (ISSN (E): 0973-9769)
75. Sathiyamoorthy V, **Sekar T** and Elango N (2015), 'Optimization of processing parameters in ECM of Automotive die tool steel using MOGA', International Journal of Mechanical Engineering Research (IJMER), Vol. 5, No. 1, pp: 9-18. (ISSN (E):2249-0019)
76. Sathiyamoorthy V, **Sekar T**, Suresh P, Vijayan R and Elango N (2015), 'Optimization of processing parameters in Electrochemical Machining of AISI 202

- using Response Surface Methodology’, Journal of Engineering Science and Technology (JEST), Vol.10, No.6, pp:780-789. (ISSN (E): 1823-4690)
77. Suresh P, Venkatesan R, **Sekar T** , Elango N (2014) ‘Optimization of intervening variables in micro-EDM of SS 316L using Genetic Algorithm and Response Surface Methodology’, Strojniški Vestnik Journal of Mechanical Engineering, Vol. 60 No.10, pp:656-664. (ISSN (E): 0039-2480)
78. Suresh P, Venkatesan R, **Sekar T** and Sathiyamoorthy V (2014), ‘Study of micro-EDM parameter of Stainless Steel 316L using Genetic algorithm’, International Journal of Engineering and Technology (IJET), Vol. 6, No. 2, pp.1065-1071. (ISSN (E): 2227-524X)
79. Loganathan V , Vijayan R and **Sekar T** (2014), ‘Mathematical modeling and Comparison on the effect of various alternative fuels on the performance of gas turbine combustion process’, IOSR Journal of Mechanical and Civil Engineering (IOSR-JMEC), Vol.11, Issue 1, pp. 86-93. (ISSN (E): 2278-1684)
80. Pasumpon M, Nallappan R, Vijayan R and **Sekar T** (2014), ‘Design and flow analysis of electromagnetic plasma arc propulsion system’, International Journal of Engineering Research, Vol.2, Issue 2, pp. 316–323. (ISSN (E): 0974-3154)
81. Srikanth M, **Sekar T** and Vijayan R (2014), ‘Performance and emission characteristics of a CI engine fuelled with Diesel 2-Ethoxy Ethyl Acetate Blends’, International Journal of Research and Innovation in Engineering Technology, (IJRIET), Vol. 1, Issue: 1, pp. 13-18. (ISSN (E): 2394–4854)
82. Sathiyamoorthy V and **Sekar T** (2014), ‘Experimental studies on improving the performance of Electrochemical machining of High Carbon High Chromium Die Steel using jet patterns’, Carbon - Science and Technology, Vol.6, Issue 1, pp. 321-329. (ISSN (E): 0974-0546)
83. Sathiyamoorthy V, **Sekar T** and Elango N (2013), ‘Experimental investigation on the effects of copper Nano particles suspended NaNO₃ electrolyte in ECM’, International Review of Mechanical Engineering (IREME), Vol.7, N.7, pp. 1523-1528. (ISSN (E): 1970-8734)
84. Yuvararasimman P, **Sekar T** , Malayalamurthi R, Balusamy T and Raja M (2010), ‘Computer aided design and analysis of air suspension system for LCV’, International Journal for Production Technology and Management (IJPTM), Vol. 1 No. 1, pp. 49-53.

85. **Sekar T** and Marappan R (2008), 'Experimental investigations into the influencing parameters of Electrochemical machining of AISI 202', Journal of Advanced Manufacturing System (JAMS), Vol. 7, Issue 2, pp. 337-343. (ISSN (E): 0219-6867)
86. **Sekar T** and Marappan R (2008), 'Investigating the Material removal rate and surface roughness in Electrochemical machining of High Carbon High Chromium Die Steel', International Journal of Engineering Research and Industrial Applications, Vol. 1, No. III, pp.121-130. (ISSN (E): 0974-1518)
87. **Sekar T** and Marappan R (2008), 'Experimental investigations into Electrochemical machining of High Carbon High Chromium Die Steel', International Journal of Applied Engineering Research, Vol. 3, No. 2, pp. 203-210 (ISSN (E):0973-4562)
88. **Sekar T** and Marappan R (2008), 'Improving Material removal rate of Electrochemical machining by using rotating tool', Journal Manufacturing Engineering, Slovakia, Vol.VIII, Issue. 2, pp. 60-63. (ISSN (E):2450-5471)
89. **Sekar T** and Marappan R (2008), 'Optimum selection of machining parameters in Electrochemical machining using Response Surface Methodology', International Journal of Manufacturing Science and Production, Vol. 9, Issue 1, pp. 11-18. (ISSN (E):2191-4184)
90. **Sekar T** and R Marappan (2007), "Experimental Studies on Effect of Tool Geometry Over Metal Removal Rate in ECM Process", Journal of Nonconventional Technologies, Romania, Issue 3, pp.107-110 (ISSN (E):2359-8654)

National Journals:

1. **Sekar T** and Marappan R (2008), 'Experimental studies on the effect of Electrolyte jet geometry over Metal removal rate in ECM processes', Manufacturing Technology Today, CMTI, Vol.7, No.5, pp.17-20.
2. **Sekar T** and Marappan R (2008), 'Intervening variables in Electrochemical machining of High Carbon High Chromium die tool steel', Journal of Manufacturing Engineering, SME, Kovilpatty, Vol.3, Issue 3, pp.150-154.

Book Chapter Authored:

1. **T.Sekar**, V.Sathiyamoorthy, K.Muthusamy, A.Sivakumar and S.Balamurugan (2021), 'Artificial Fish Swarm Algorithm Driven Optimization for Copper-Nano Particles Suspended Sodium Nitrate Electrolyte Enabled ECM on Die Tool Steel', Springer Futuristic Trends in Intelligent Manufacturing, Book Chapter, pp.47-50 ISBN: 978-3-030-70008-9.

2. M.Vijay, **T.Sekar**, N.Muthukumaran and K.Vijayakumar (2021), 'Investigations on Electrochemical Discharge Machining of Al₂O₃ Ceramics" Intelligent Manufacturing and Energy Sustainability, Smart Innovation, Systems and Technologies, Springer, Vol.213, pp.33 - 43 ISBN Online: 978-981-33-4443-3.

Books Authored

1. Sivakumar Aburpa Avanachari, **Sekar Tamilperuvalathan** and Sathiyamoorthy Varatharajan (2019), 'Electrochemical Machining of AISI202 using mixed Electrolytes', LAP LAMBERT Academic Publishing, Germany, ISBN: 978-613-9-45643-7.
2. Sathiyamoorthy Varatharajan, **Sekar Tamilperuvalathan** and Sivakumar Aburpa Avanachari (2018), Improving the MRR of ECM using Nano particles suspended Electrolytes', LAP LAMBERT Academic Publishing, Germany, ISBN: 976-613-9-45060-2.
3. **Sekar Tamilperuvalathan** (2017), Experimental Investigation on Electrochemical Machining, LAP LAMBERT Academic Publishing, Germany, ISBN-13: 978-3330335400.
4. **T.Sekar** (2000), 'CAD/CAM Short Answer-Questions Bank', Ramalinga Publications, Sothupakkam, India.

Academic and Professional Achievements:

1. Board of Governing Council Member for Nehru Institute of Technology, Coimbatore & United Infotech Institute of Technology, Coimbatore.
2. GCT, Officer-in-charge of Examinations from April, 2024 onwards
3. Coordinator of Intellectual Property Rights Cell, GCT, Coimbatore.
4. **Government of Tamil Nadu nominee** for IQAC of Coimbatore Institute of Technology, Coimbatore.
5. **Anna University Nominee** for the BoS of Chenduran College of Engineering, Pudukottai, Al-Ameen Engineering College (Autonomous), Erode, Sri Krishna College of Technology, Coimbatore and The Kaveri Engineering College, Mecheri and Ganesh College of Engineering, Salem.
6. Chennai Anna University Affiliation –Inspection Committee Member/Chairman for granting provisional affiliation.
7. **Head** of M.E- Manufacturing Engineering, Government College of Technology, Coimbatore and provided 100 % placement for the batch of 2020-22 due to initiating sincere efforts by the department.

8. **Government of Tamil Nadu Nominee** for CSI College of Engineering, The Nilgiris, Asian College of Engineering and Technology, Coimbatore and Sri Narayanaguru & Sankara Polytechnic Colleges, Coimbatore.
9. Initiator of '**Business Incubator**' cell (Central Government of India Scheme) at TPGIT, Vellore, India.
10. Successfully organized a **National Level Conference on Emerging Technologies in Mechanical and Manufacturing Engineering (e-TimmE2K17)** as a Coordinator at the newly established Government College of Engineering, Dharmapuri on 22.02.2017.
11. Acted as a **Zonal Officer** of Zone –VII, (*Anna University*), GCE, Dharmapuri from 2016 to 2017.
12. **TEQIP Phase II –Officer**, World Bank aided Technical Education Quality Improvement Program (Sanctioned Amount of **Rs. 12.5 Cr**), Government College of Engineering, Salem from February, 2014 to August, 2014 and completed the Data/Performance audit/stock verification activities very successfully. Also steps have been initiated to offer One Credit Courses through the established Industry-Institute Interaction Cell.
13. **Program Officer- NSS Unit –IV** of GCE, Salem from April, 2002 to November, 2010 and planted/maintained around 5,000 saplings/trees at the premises of GCE, Salem and the unit's adopted villages.
14. Production and Thermal Engineering Labs commissioned at GCE, Dharmapuri as a **Head of the Department**.
15. Served as a **Placement & Training Officer** at Government College of Engineering, Salem, Tamil Nadu, India between 2009 and 2010. (*Inception in the year 1966 and it was started as the 'Third Government Engineering, College' in Tamil Nadu*).
16. **Proactive Reviewer for Q1/Q2 Journals (Elsevier, Springer etc.)**
17. **Former Editorial Board Advisory Member** of Journal of Engineering Science and Technology, Taylor's University, Malaysia.
18. Served as a **Hostel Superintendent**, Officer in-charge for Vehicle and Library at GCE, Dharmapuri from 2015 to 2017.
19. **Board of Studies Member** for GCE, Salem from 2013-2014 & in GCT, Coimbatore from 2018 to till date.
20. **BoS -Anna University nominee** to KSR College of Technology, Tiruchengode from 2014 to 2016.
21. **Academic Auditor** and Department Advisory Board Member at GCT, Coimbatore.

Additional Administrative Experience at DoTE

Worked at Head Quarters i.e., at Directorate of Technical Education, Chennai for Tamil Nadu Engineering College Admissions-2019 - Seat Matrix related activities and seeking for First Generation Graduate Tuition Fee Waiver Scheme (FGG). Students' Admission Approval, Post Matric Scholarship Scheme related activities.

I strongly believe that quality education should be accessible to every deserving student irrespective of social or economic background. In this regard, I had the privilege of contributing to the implementation of the Government of Tamil Nadu's landmark 7.5% reservation scheme for Government School students seeking admission to professional degree programmes. During my tenure at the Directorate of Technical Education, Chennai, I worked closely with the committee headed by Hon'ble Chief Justice (Retd.) Shri D. Murugesan, supporting efforts that opened the doors of higher education to thousands of first-generation learners and economically disadvantaged students.

Funded Project- completed Details:

1. Successfully completed the TEQIP-III sponsored Research & Development Project on **‘Experimental Investigations on Electrochemical Discharge Machining (ECDM) of Bio-Ceramics’** at P.G-Manufacturing Engineering, Department of Mechanical Engineering, Government College of Technology, Coimbatore as the Principal Investigator **(T.Sekar)** vide Office Proceedings No.1096/TEQIP-III/R & D/2020, Dated. 18.09.2020, Sanctioned Amount: **Rs. 1,90,941 /-**

Patent Details:

1. TWELVE number of Patents were filed. Among the twelve, **TEN** have been granted and remaining are under processing (Design Patent Nos. 408368-001, 20.03.2024, 411101-001, 10.05.2024, 411325-001, 16.05.2024, 418204-001, 28.05.2024, 420065-001, 14.06.2024, 419712, 12.06.2024, 419047-001, 05.06.2024, 408376-001, 22.02.2024, 436796-001, 08.11.2024 & 472903-001, 09.09.2025) (https://drive.google.com/file/d/1JZB0Xcu0vQdkl37BGZXONnMHUd_UGTTp/view?usp=sharing)
2. A patent was filed on “Center stand Mechanism for Two-Wheelers offering enhanced convenience and Ease of use”, Filing date: 2024-12-28. (T.Sekar as an APPLICANT).
3. A patent was filed on “Inclined, Spiral Electrolyte Jets in a Rotating Tool Improve Electrochemical Machining Performance”, 1049/CHE/2007 A, Filing date: 2007-05-18, Publication date: 2008-12-05. (T.Sekar as the Principal Investigator). However, taken-up the foreign assignment during 2010-12, unable to proceed further more.

Memberships

1. Senior Individual Member of INAE
2. Fellowship of the Indian Institution of Engineers (FIE)
3. Life Member in Indian Society for Technical Education (MISTE)
4. Life Member in Indian Institution of Production Engineers (MIIPE)
5. Life Member in Material Research Society of India (MMRSI)
6. Former Member (Chartered Engineer), Institution of Mechanical Engineers (IMechE) from 2016 to 2022

Personal Details:

1. DOB : 12.07.1975
2. Native Place & District: Tittagudi (606106) & Cuddalore, Tamil Nadu, India
3. Alumnus of PSG College of Technology, Coimbatore (U.G), Annamalai University, Chidambaram (P.G- Medalist), Anna University, Chennai (Ph.D) & Periyar University, Salem (M.B.A- Human Resources) & NEBOSH
4. A Valid Indian Passport Holder (Until 2028)
5. Proudly say that I am a student of Government Higher Secondary School, Tittagudi.

Coimbatore
20.07.2026

Sd/.
T.Sekar